Solar Potential (terrain illumination)

2024-12-21 · 1,278.8 km²

0.336	22.1 %	70.0°	180.0°
Mean relative irradiance	Cells facing away from the sun	Sun zenith angle	Sun azimuth

Executive summary

- At 12 h on 2024-12-21 the sun is 70.0° from the zenith (azimuth 180°). Mean relative irradiance on the terrain is 0.34 (1 = sun directly overhead of a surface facing it).
- 22.1 % of cells face away from the sun at that moment.

> Single-instant relative factor; no terrain shadowing, no atmosphere.

Study area and inputs

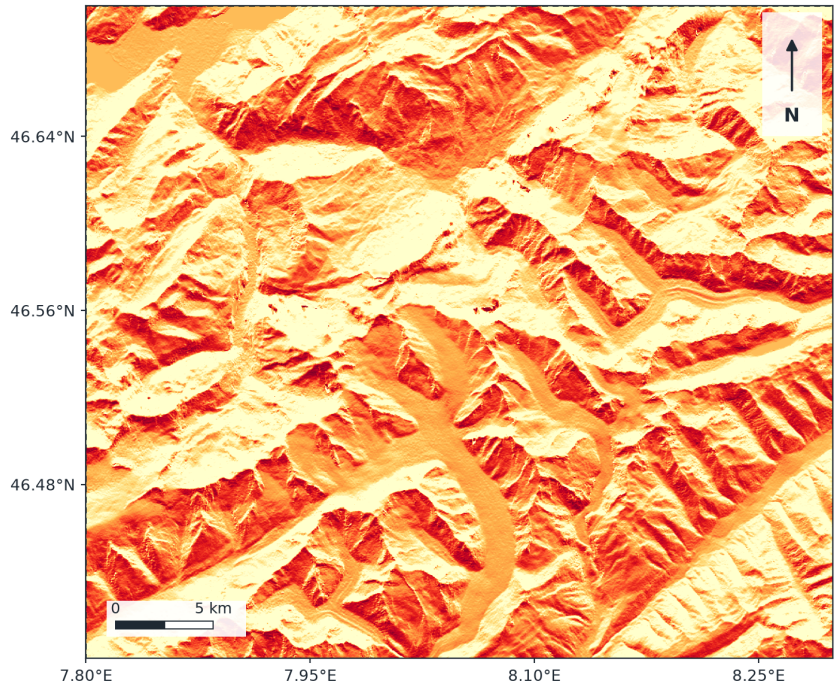
Area of interest (west, south, east, north)	7.8000, 46.4000, 8.3000, 46.7000
Centre	46.5500°, 8.0500°
Area of interest	1,278.78 km²
Period	2024-12-21
Output grid	1,800 × 1,080 pixels, EPSG:4326
Raster extent area	1,278.78 km²

Inputs: a digital elevation model (srtm1arc) from OpenTopography.

Results

Solar Potential (terrain illumination)

Relative clear-sky irradiance (0-1) · 2024-12-21 · AOI 7.8000, 46.4000, 8.3000, 46.7000



relative irradiance

0.00 0.25 0.50 0.75 1.00

Mean	**0.336**
Median	**0.342**
Min / Max	**0 / 1**

Key figures

Mean relative irradiance
0.336

Cells facing away from the sun
22.1 %

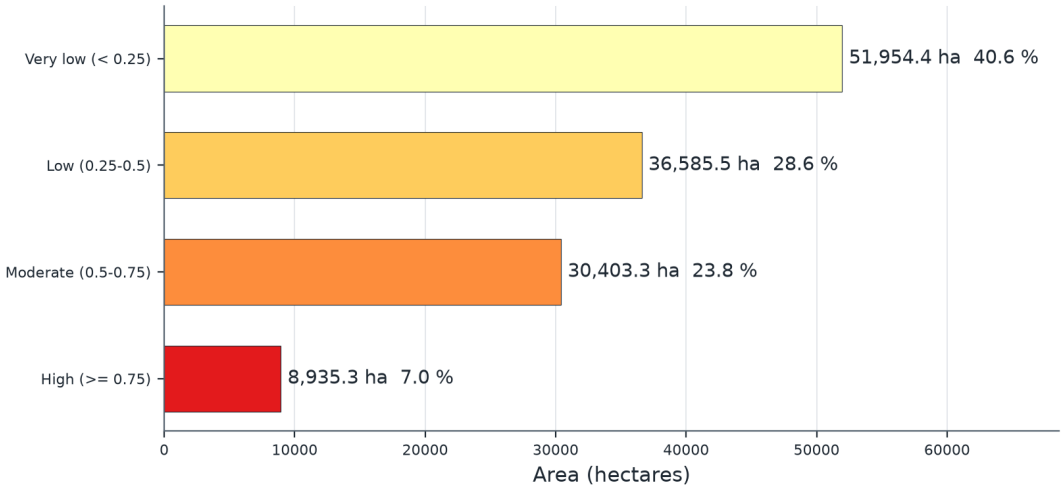
Sun zenith angle
70.0°

Sun azimuth
180.0°

CRS: EPSG:4326 · Areas measured from the output raster · No basemap · PyGeoVision 2.2.0 · 2026-10-17 19:17 UTC

Map: Solar Potential (terrain illumination)

Area by class: Relative clear-sky irradiance (0-1)



Summary chart

Class areas: Relative clear-sky irradiance (0-1)

Class	Value	Pixels	Area (ha)	Share of valid area	Share of AOI
Very low (< 0.25)	0	789,910	51,954.4	40.6 %	40.6 %
Low (0.25-0.5)	1	556,162	36,585.5	28.6 %	28.6 %
Moderate (0.5-0.75)	2	462,110	30,403.3	23.8 %	23.8 %
High (>= 0.75)	3	135,818	8,935.3	7.0 %	7.0 %

Valid area 127,878.5 ha of 127,878.5 ha (1,944,000 of 1,944,000 pixels). Areas are measured from the output raster.

Key statistics (reported by the pipeline)

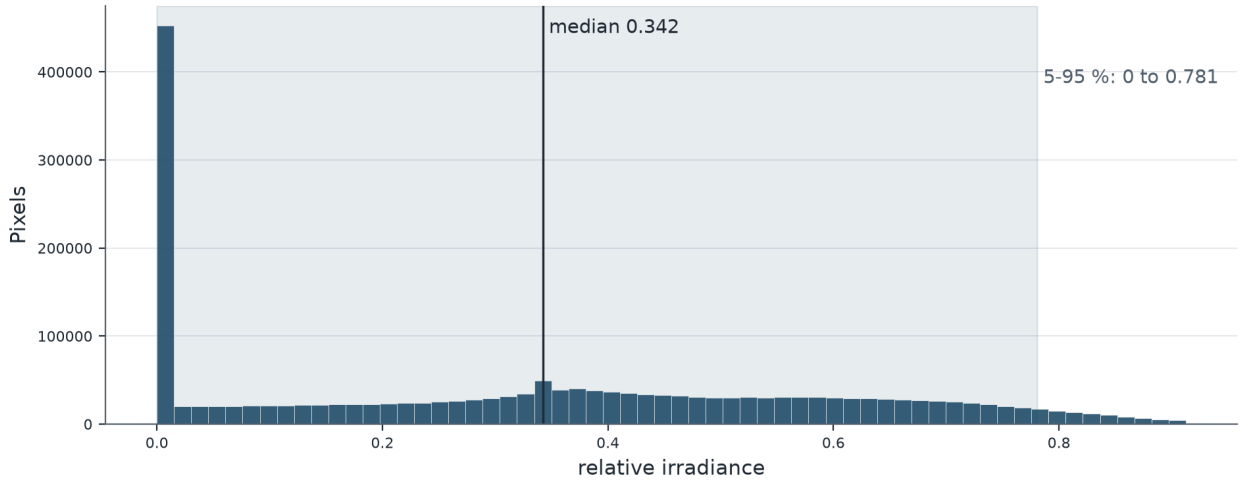
Measure	Value
Mean relative irradiance	0.336
Cells facing away from the sun	22.1 %
Sun zenith angle	70.0°
Sun azimuth	180.0°
Date	2024-12-21
Hour (solar time)	12.0
DEM product	srtm1arc

Value statistics: Relative clear-sky irradiance (0-1)

Statistic	Value
Minimum	0.000
5th percentile	0.000
25th percentile	0.040
Median	0.342
Mean	0.336
75th percentile	0.556
95th percentile	0.781
Maximum	1.000
Standard deviation	0.269

Over 1,944,000 valid pixels.

Distribution: Relative clear-sky irradiance (0-1)



Distribution of values

Method and parameters

How much does terrain slope and aspect help or hinder sunlight at a given date and hour?

Slope and aspect come from the DEM (Horn's method). The sun position follows the standard declination and hour-angle equations.

The output is the cosine of the sun's incidence angle on each cell (0-1), for one instant.

Pipeline	solar_potential
Area of interest	7.8000, 46.4000, 8.3000, 46.7000
cloud_mask	True
date	2024-12-21
hour	12

Interpretation and limitations

Limitations

- A single-instant relative factor: no terrain shadowing by neighbouring ridges, no atmosphere, no clouds, and not kWh/m2.
- Annual yield needs a time-integrated model with measured or modelled irradiance.

Deliverables

Files

File	Type	Contents
alps_winter_sun/solar_potential.tif	GeoTIFF raster	Relative clear-sky irradiance (0-1)
alps_winter_sun/deliverables/solar_potential_report.pdf	PDF report	Complete analysis report
alps_winter_sun/deliverables/solar_potential_report.xlsx	Excel workbook	Complete analysis report
alps_winter_sun/deliverables/solar_potential_report.docx	Word report	Complete analysis report
alps_winter_sun/deliverables/maps/solar_potential_map.png	PNG image	Map or chart for slides and documents
alps_winter_sun/deliverables/maps/solar_potential_chart.png	PNG image	Map or chart for slides and documents
alps_winter_sun/deliverables/maps/solar_potential_histogram.png	PNG image	Map or chart for slides and documents
alps_winter_sun/deliverables/vectors/solar_potential.gpkg	GeoPackage (all layers)	Vector features with attributes
alps_winter_sun/deliverables/vectors/geojson/solar_potential_aoi.geojson	GeoJSON, EPSG:4326	Vector features with attributes
alps_winter_sun/deliverables/vectors/geojson/solar_potential_irradiance_classes.geojson	GeoJSON, EPSG:4326	Vector features with attributes
run_summary.json	JSON	Machine-readable record of this run

Vector layers

Layer	Features	Total area (ha)	Note
irradiance_classes	26,301	127,878.5	

Vector layers made from rasters carry class_id, class_name, color_hex, area_ha, area_m2 (equal-area projection), pct_aoi, cx, cy, pipeline and period; layers a pipeline wrote itself (buildings, roads, streams, fire points) keep their own attributes. Coordinates are WGS 84 (EPSG:4326). The GeoPackage also holds the 'aoi' outline and, for raster-derived layers, a dissolved '<layer>_classes' summary.

References

- Duffie, J.A. & Beckman, W.A. Solar Engineering of Thermal Processes (Wiley); Spencer, J.W. (1971). Fourier series representation of the position of the sun. Search 2(5), 172.
- Horn, B.K.P. (1981). Hill shading and the reflectance map. Proceedings of the IEEE 69(1), 14-47.